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Department of Electronics & Communication Engineering | GATE Preparation Library

Analog Circuits — Official GATE ECE Syllabus

Paper Code: EC • Section 4: Analog Circuits (8 - 11 Marks)

Weightage: 8 - 11 Marks • **Subject Code:** AC • **Key Focus:** BJT/MOSFET Amplifiers, Op-Amps, Feedback

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: DIODE CIRCUITS & POWER SUPPLIES

- Small-signal diode model: Dynamic resistance (rd = eta * Vt / ID), diffusion and depletion capacitance effects.
- Clipping circuits: Single-ended and double-ended diode clippers with DC reference voltages.
- Clamping circuits (DC restorers): Peak clampers, clamping theorem (area of forward current = area of reverse current).
- Rectifier circuits: Half-wave rectifier, center-tapped full-wave rectifier, bridge rectifier.
- Performance parameters: Peak Inverse Voltage (PIV), ripple factor (gamma = 1.21 for HWR, gamma = 0.482 for FWR), rectification efficiency (40.6% for HWR, 81.2% for FWR), transformer utilization factor (TUF).
- Capacitor filters and LC filters: Ripple factor approximation (gamma = 1 / (2*sqrt(3)*f*R_L*C)).
- Voltage regulation using Zener diodes: Minimum and maximum Zener current constraints, load and line regulation limits.

SECTION: BJT AND MOSFET AMPLIFIERS & BIASING

- BJT biasing and DC operating point (Q-point): Fixed bias, collector-to-base feedback bias, self-bias (voltage divider bias).
- Stability factors: S(Ico), S(Vbe), S(beta). Thermal runaway condition: dPc/dT < 1/theta.
- MOSFET biasing circuits: Voltage divider bias, current source active biasing.
- Small-signal low-frequency models: Hybrid-pi and T-models for BJTs and MOSFETs. Transconductance parameters: gm = Ic/Vt for BJT, gm = sqrt(2*mu*Cox*(W/L)*Id) = 2*Id/(Vgs - Vt) for MOSFET.
- Single-stage BJT amplifier configurations: Common Emitter (CE) with and without emitter degeneration, Common Collector (CC / Emitter Follower), Common Base (CB).
- Single-stage MOSFET amplifier configurations: Common Source (CS), Common Drain (CD / Source Follower), Common Gate (CG).
- Performance comparison: Voltage gain (Av), current gain (Ai), input impedance (Rin), output impedance (Rout), and gain-bandwidth product.
- High-frequency small-signal response: Miller's theorem and Miller effect on input capacitance (Cin = C_base-emitter + C_base-collector*(1 - Av)).
- Short-circuit cutoff frequency (f_beta and f_T) for BJT, Unity-gain frequency for MOSFET.

SECTION: DIFFERENTIAL, MULTI-STAGE & FEEDBACK AMPLIFIERS

- Multi-stage amplifiers: RC-coupled, direct-coupled, and transformer-coupled amplifiers. Overall gain, bandwidth shrinkage in cascade.
- Differential pair amplifiers (BJT and MOS pairs): Large-signal characteristics, small-signal differential gain (Ad), common-mode gain (Acm).
- Common Mode Rejection Ratio: CMRR = |Ad / Acm|, CMRR in dB. Input common-mode range (ICMR), differential input resistance.

• Active loads and current mirrors: Simple two-transistor current mirror, Widlar current source, Wilson current source, Cascode current mirror.

• Feedback amplifier principles: Negative feedback concept, loop gain (T = A*beta), return ratio.

• Four feedback topologies: Voltage-Series (Series-Shunt), Voltage-Shunt (Shunt-Shunt), Current-Series (Series-Series), Current-Shunt (Shunt-Series).

• Effect of negative feedback on circuit performance: Gain desensitization (dA_f/A_f = (1/(1+A*beta))*(dA/A)), bandwidth extension (BW_f = BW*(1 + A*beta)), reduction in non-linear distortion, noise reduction, and input/output resistance transformations.

SECTION: OPERATIONAL AMPLIFIERS (OP-AMPS) & APPLICATIONS

• Ideal vs. practical operational amplifier characteristics: Open-loop gain (Aol -> infinity), input resistance (Rin -> infinity), output resistance (Rout -> 0), bandwidth, offset voltage, bias currents, offset current.

• Slew Rate (SR = dVo/dt_max = 2*pi*f_max*Vom), Full-power bandwidth (fpbw).

• Virtual ground and virtual short concepts in negative feedback configurations.

• Linear applications: Inverting and non-inverting amplifiers, voltage follower (buffer), summing amplifier, difference amplifier, instrumentation amplifier (3 op-amp structure, CMRR adjustment).

• Integrator (frequency response, DC stabilization with shunt resistor), Differentiator (high-frequency noise reduction with series input resistor).

• Logarithmic and anti-logarithmic (exponential) amplifiers using diodes/BJTs in feedback, analog multipliers.

• Non-linear applications: Voltage comparators, open-loop saturation, zero-crossing detectors.

• Schmitt trigger: Inverting and non-inverting Schmitt triggers, upper threshold voltage (VUTP), lower threshold voltage (VLTP), hysteresis voltage (Vh = VUTP - VLTP).

• Precision rectifiers: Super-diode, precision half-wave and full-wave rectifiers, peak detectors, sample-and-hold circuits.

SECTION: ACTIVE FILTERS, OSCILLATORS & 555 TIMERS

• Active filters: First-order and second-order low-pass, high-pass, band-pass, and band-reject (notch) Butterworth filters. Sallen-Key topology.

• Oscillator principles: Positive feedback, Barkhausen criterion for sustained oscillations (|A*beta| = 1 and phase shift = 0 or 360 degrees).

• RC oscillators: RC phase-shift oscillator (f0 = 1 / (2*pi*R*C*sqrt(6)), BJT/Op-Amp gain requirement Av >= 29), Wien-bridge oscillator (f0 = 1 / (2*pi*R*C), gain requirement Av >= 3).

• LC tuned oscillators: Hartley oscillator (tapped inductor), Colpitts oscillator (tapped capacitor, f0 = 1 / (2*pi*sqrt(L*C_eq))), Crystal oscillators (piezoelectric effect, series and parallel resonance, high Q stability).

• IC 555 Timer architecture: Internal block diagram (resistive divider, two comparators, SR flip-flop, discharge transistor, output stage).

• 555 Timer in Monostable multivibrator mode: Pulse width (T = 1.1 * R * C).

• 555 Timer in Astable multivibrator mode: Charge and discharge intervals, oscillation frequency (f = 1.44 / ((RA + 2*RB)*C)), duty cycle calculation.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

• Microelectronic Circuits — Adel S. Sedra & Kenneth C. Smith (Oxford University Press)
• Electronic Devices and Circuit Theory — Robert L. Boylestad & Louis Nashelsky
• Op-Amps and Linear Integrated Circuits — Ramakant A. Gayakwad

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

• Virtual ground is valid ONLY when negative feedback is established and op-amp is NOT saturated.
• Op-Amp Slew Rate limitation: If input frequency f > SR / (2*pi*V_peak), output waveform distorts from sinusoid into triangular wave.
• Schmitt trigger trip points: Always write KCL at the non-inverting terminal (+) to determine VUTP and VLTP.